

## What Is A Model?

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Architects and designers for example, make prototypes of the buildings they design to preview how they would look once finished. Typically, a model is a small-scale version of the real thing. Such a representation could either be physical (like a miniature building made with glass and board) or virtual (a computer generated 3D image using tools such as CADD—Computer Aided Design and Drafting).

SketchUp is a really flexible, simple and easy-to-use tool that any layman—with a bit of practice—can use to make 3-Dimensional models. With SketchUp, now everyone can make a model without the technical hurdles and learning-related delays involved in getting the hang of other tools in the market, like AutoCAD, 3ds Max, etc.

## 2.1 The Drawing Axes

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As soon as you open SketchUp, what you see is the *drawing area*, which is basically a space provided to you to create your model.

If you're really bright and didn't bunk classes, you'll remember coordinate geometry in school. A coordinate system is basically a method of locating points or positions by assigning numbers to them. A three dimensional coordinate system—like the one in SketchUp—has 3 axes and is used to locate a point or object in space. This helps to represent objects in 3D (length, breadth and depth/height) on a 2D surface (a flat screen or a sheet of paper).

SketchUp provides three drawing axes: X, Y and Z. Each axis represents a particular dimension and these three axes help to identify the 3D space of the drawing area.

**X-axis:** The X axis is represented by a solid red line, with the minus X part of the axis represented by a dotted red line.

**Y-axis:** The Y axis is the green line (solid for plus and dotted for minus)

**Z-axis:** The Z axis is the blue line you see. Plus or minus Z (above and below the ground plane) are represented by solid blue and dotted blue lines, respectively.

The point of intersection is called the origin. This is represented by a black circle in SketchUp. Each axis has its negative coordinates